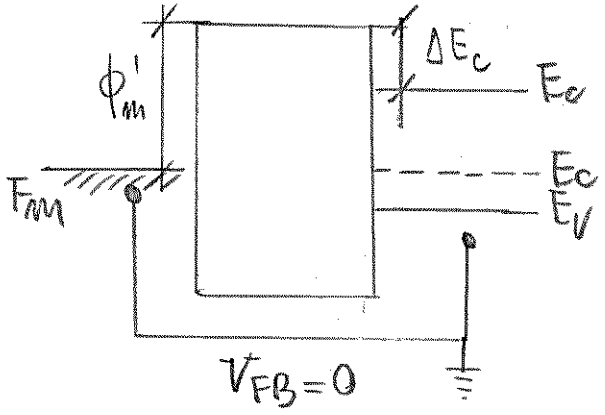
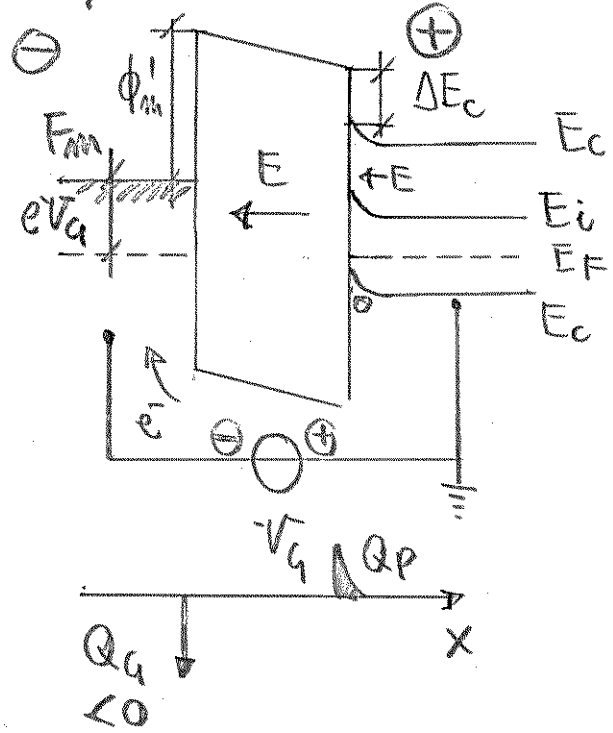
MOS Capacitor

$F_m = E_F$ or $\phi_m = \phi_s \rightarrow$ Flat bands



Accumulation Regime

$V_G < 0$



$\vec{E} - \nabla V$

$E_s(x) = \frac{1}{q}\frac{dE_c}{dx} < 0$

Gauss th.

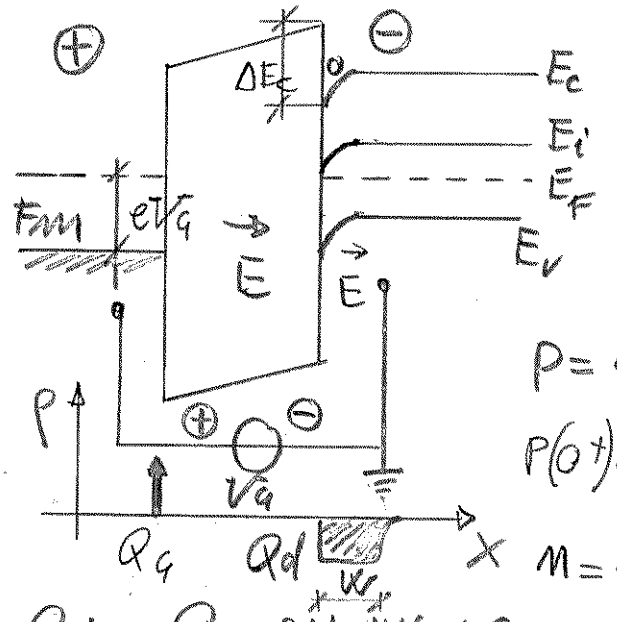
$\frac{E_i^{(0)}}{\varepsilon_s} = \frac{E_s^{(0)}}{\varepsilon_i}$ $[x=0]$

$E_i^{(0)} = \underbrace{\frac{\varepsilon_s}{\varepsilon_i}}_{3} E_s^{(0)}$ $\begin{bmatrix}\varepsilon_s = 12 \\ \varepsilon_i = 4\end{bmatrix}$

$p = n_i \exp\left(\frac{E_i - E_F}{KT}\right) > n_a$

$Q_p = -Q_G$

Depletion Regime $V_G > 0$



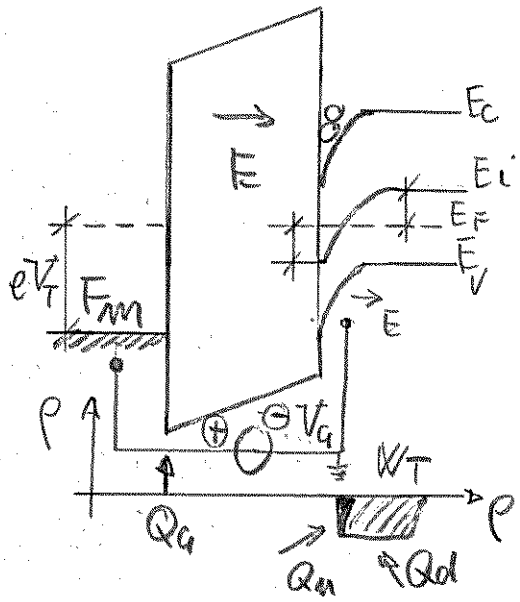
$p = n_i \exp\left(\frac{E_i - E_F}{KT}\right)$

$p(0^+) \ll n_a$

$n = n_i \exp\left(\frac{E_F - E_i}{KT}\right)$

$Q_d = -Q_G = q n_a W < 0$

Threshold for strong inversion $V_G = V_T$



$n = n_i \exp\left(\frac{E_F - E_i}{KT}\right)$

$[E_F - E_i]_{x=0^+} = [E_i - E_F]_{bulk}$

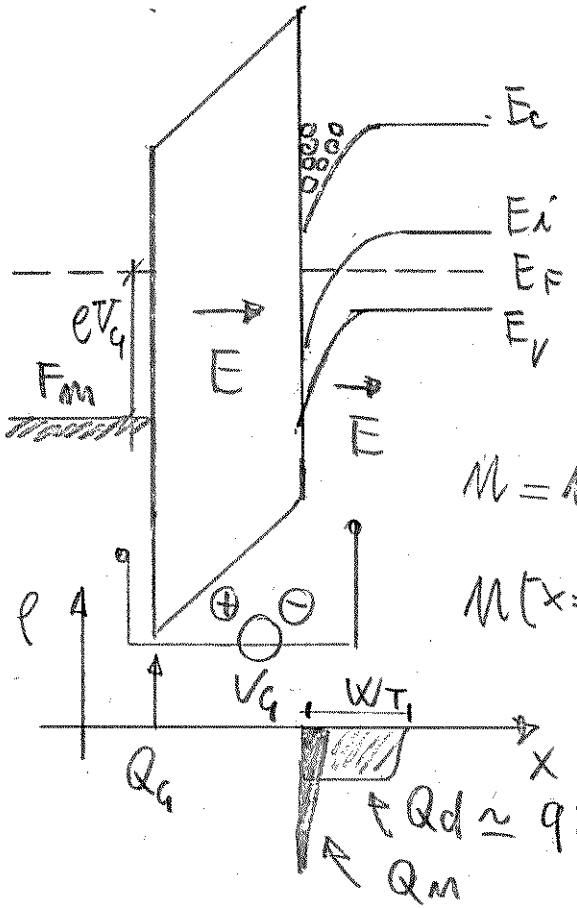
$n[x=0^+] = p[bulk]$

$p(0^+) \ll n_a$

$Q_G = -Q_d - Q_n$

Strong Inversion

$V_G >> 0$



$$[E_F - E_i]_{x=0^+} > [E_i - E_F]_{bulk}$$

$$n = n_i \exp\left(\frac{E_F - E_i}{KT}\right)$$

$$n[x=0^+] >> N_a$$

$$\begin{cases} J_n = \mu_e n \dfrac{dF_n}{dx} = 0 \\ J_p = \mu_e p \dfrac{dF_p}{dx} = 0 \end{cases}$$

$\hookrightarrow F_n = F_p$

$Q_n >> Q_d \qquad Q_G = -Q_d - Q_n$

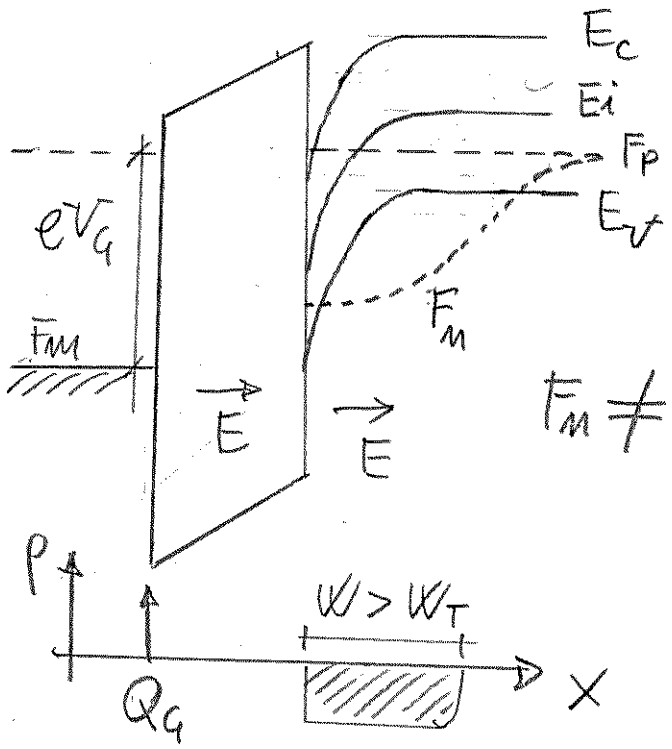
The thickness W of the depletion layer is almost constant to W_T (Threshold)

In fact:

$\Delta Q_d \propto \Delta W \qquad \Delta Q_n \propto \exp(\Delta W)$

Deep Depletion $V_G >> V_T$

Non equilibrium



$F_n \neq F_p$

$Q_d = q N_a W \qquad Q_n = 0$

$Q_G = Q_d$

$$np = n_i^2 \exp\left(\frac{F_n - F_p}{KT}\right) << n_i^2$$

charge generation g

After a long time $\rightarrow$

$\rightarrow$ strong inversion